

Deribit ATM Implied Volatility as a Forward Realized-Volatility Forecast

Options-implied volatility as a venue-agnostic risk & sizing conditioner for executable Bitcoin strategies

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ABSTRACT

We document the one positive, reproducible result from 19V Capital's Bitcoin / Deribit-options research program: the month-start 30-day at-the-money implied volatility (ATM IV) of Deribit Bitcoin options is a strong, era-stable forecast of Bitcoin's realized volatility over the subsequent 21 trading days. In a pre-registered, seed-reproducible study, ATM IV attains a Spearman rank information coefficient of **+0.66** (Newey-West $t = +6.69$, $n = 60$), clears the strictest multiple-testing bar across a 60-signal family, holds its sign across every observable market era, and statistically subsumes trailing realized volatility — the obvious competing predictor adds essentially nothing once implied vol is included.

A directional hypothesis — that the surface predicts the *sign* of forward returns — was pre-registered and rejected. A companion application backtest shows that sizing a monthly volatility-targeted Bitcoin exposure on ATM IV tracks the volatility target materially tighter, trades cheaper, and roughly halves drawdown relative to an unsized position. The signal's value is therefore as a **volatility / risk conditioner** — position sizing, regime-gating, vol-targeting — rather than as a directional alpha source. All results are research-grade; the binding out-of-sample validation pass remains outstanding.

Keywords: implied volatility; realized-volatility forecasting; Bitcoin; Deribit options; volatility targeting; information coefficient; position sizing; options-implied information.

Status: research fidelity (not yet binding-validated) · Pre-registered and reproducible (seed 20260611)

Two underlying studies: a signal study (verdict ITERATE) and a vol-conditioner application backtest (verdict HANDOFF)

HEADLINE RESULT

The month-start **30-day at-the-money implied volatility** of Deribit Bitcoin options forecasts **Bitcoin's realized volatility over the next 21 trading days**. Rank correlation (Spearman IC) **+0.66**; Newey-West t-statistic **+6.69** (n=60); clears the strictest multiple-testing bar across a 60-signal family; **stable across every observable market era**; and it **statistically absorbs trailing realized volatility** (the obvious competing predictor adds essentially nothing once implied vol is included). Directional use — predicting the *sign* of returns — was tested and **failed**. The signal's value is therefore as a **volatility/risk conditioner** (sizing, regime-gating, vol-targeting), not as a directional alpha source.

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1. Executive summary

This dossier documents the one genuinely positive, reproducible result from 19V Capital's Bitcoin / Deribit-options research effort. Two studies were run, both at research fidelity, both pre-registered (hypotheses and decision rules fixed before results were seen), both seeded for exact reproduction.

- **Study 1 (signal study).** Establishes the core fact: Deribit month-start 30-day ATM implied volatility (IV) is a strong, era-stable forecast of forward realized volatility (RV), and it subsumes trailing RV. The directional hypothesis (does the surface predict return *direction*?) was tested and killed, as expected. Verdict: **ITERATE** — advance the IV→RV finding as an input; do not treat it as a standalone strategy.
- **Study 2 (application backtest).** Takes the forecast and tests whether it *sizes a position* better than the conventional trailing-RV estimate, by running a monthly volatility-targeted long-Bitcoin exposure. IV-based sizing tracks the volatility target materially tighter, runs cheaper, and roughly halves drawdown versus an unsized position. Verdict: **HANDOFF** — ready for the full binding validation pass.

What it is, in one line: a high-quality, literature-consistent *volatility forecaster* for Bitcoin — useful as a risk/sizing conditioner for executable strategies, not as a buy/sell signal. **What remains:** the binding out-of-sample validation pass (Section 14) has not yet been run; all numbers here are research-grade.

2. Context & the question

Bitcoin is traded by 19V on listed CME futures; an associated options market (Deribit) prices implied volatility across strikes and maturities. A standing question in the research program was whether

information in the Deribit options *surface* — implied vol level, term-structure slope, skew, put/call open interest — carries forward-looking signal about Bitcoin that could improve an executable futures strategy.

Two distinct claims were separated up front, because they have very different prior probabilities and very different value:

- **H1 — volatility forecasting (primary).** Does month-start surface information predict *how much* Bitcoin will move (realized volatility) over the following weeks? Prior: plausible — implied vol is, by construction, the market's forward vol expectation, and this relationship is one of the most replicated facts in empirical finance.
- **H2 — directional forecasting (secondary).** Does it predict *which way* Bitcoin will move (signed returns)? Prior: low — options flow is largely direction-agnostic, and any implied-minus-realized vol gap is known to be a risk premium, not a directional forecast.

The execution context matters and shapes how the result can be used. 19V's executable account enforces intraday-flat discipline (no overnight or multi-day holds) and tight real-time loss limits. A monthly-held position is therefore *not directly executable* on that account; a volatility/sizing conditioner that improves an intraday strategy *is*. This is why the deliverable is framed as a conditioner, and why a venue-agnostic, reusable forecast is strategically valuable: it survives a change of execution venue.

3. Definitions & glossary

Every term and statistic used in this document, defined so it stands alone.

Term	Definition
Implied volatility (IV)	The volatility figure implied by an option's market price — the market's forward-looking estimate of how much the underlying will move, annualized. "ATM" = at-the-money (strike near spot). "30-day" = constant-maturity 30 calendar days.
Realized volatility (RV)	The volatility actually observed after the fact, computed from price movements. "Forward 21d RV" = the RV realized over the 21 trading days <i>following</i> the observation.
Trailing RV	RV measured over a window <i>ending before</i> the observation (here, the prior 21 trading days). The standard backward-looking benchmark forecast.
Spearman IC	Information Coefficient = the rank correlation between the signal and the outcome. Ranges -1 to +1; +0.66 means a strong, monotone positive relationship robust to outliers (ranks, not raw values).
Newey-West (NW) t-stat	A t-statistic computed with a HAC (heteroskedasticity- and autocorrelation-consistent) variance estimator. It corrects standard errors for overlapping/serially-correlated observations, which would otherwise overstate significance. "NW(2)" = 2 lags. $ t \geq \sim 2$ is conventionally "significant."
Bonferroni correction	A conservative multiple-testing adjustment: if you test m signals, raise the significance bar so the chance of <i>any</i> false positive stays controlled. Here $m=60$ raises the bar to $ t \geq 3.2$.
Block bootstrap	A resampling method that draws contiguous blocks of data (preserving autocorrelation) to build a confidence interval. "CI excludes zero" = the effect is unlikely to be noise. Here: 6-month blocks, 2000 resamples.
Era stability	Whether the effect holds across distinct market regimes rather than concentrating in one period. Tested by splitting the sample (here at the Jan-2024 spot-ETF launch) and requiring same sign and independent significance in each part.
Variance risk premium (VRP)	The systematic gap between implied and subsequently realized variance — compensation paid for bearing volatility risk. It explains why IV forecasts vol but not direction.
QLIKE	A volatility-forecast loss function, $r/f - \ln(r/f) - 1$ (r = realized variance, f = forecast). It penalizes under-forecasting volatility harder than over-forecasting — the appropriate loss for risk sizing. Lower is better.
Diebold-Mariano (DM) test	A formal test of whether one forecast's loss is significantly lower than another's. A negative DM statistic vs a benchmark means the candidate forecast wins.
Pre-registration	Fixing the hypotheses, signal list, sample, and decision rules <i>in writing before</i> computing results — the discipline that prevents fitting a story to noise.
Research fidelity vs binding	"Research fidelity" = a first-pass study sufficient to decide whether to invest more. "Binding" = the full out-of-sample / walk-forward / selection-bias validation required before any capital decision.

4. Signal specification (exact formulas)

All features are computed from a single monthly snapshot — the earliest complete hourly Deribit options snapshot on the 1st of each month (typically 01:00 UTC, chosen as the first snapshot with quote count $\geq$

50% of that day's maximum). Every input is lookahead-safe (see Section 5).

Input	Exact construction
atm_iv_30 (headline feature)	Per expiry: ATM mark-IV by linear interpolation of mark-IV at delta = +0.50 (call) and -0.50 (put), averaged (filters: ≥ 3 quotes/side, delta in [0.03, 0.97], time-to-expiry 2–200 days, mark-IV in (1, 500)). Constant-maturity 30-day value by linear interpolation <i>in total variance</i> ($IV^2 \cdot T$) across the two bracketing expiries; flat extrapolation outside the listed range (flagged). Also computed at 7- and 90-day maturities.
Target: fwd_rv_h	<pre>fwd_rv_h = sqrt(252 * mean[daily 5-minute realized variance] over trading days anchor+1 ... anchor+h)</pre> where the anchor = the first Bitcoin futures trading day on/after the 1st. Horizons $h \in \{5, 10, 21\}$ trading days. Daily realized variance = sum of squared 5-minute log returns.
Target: fwd_ret_h	Log close-to-close return over h trading days from the anchor (the directional target for H2).
rv_trail_30 (benchmark)	<pre>sqrt(252 * mean[daily 5-min realized variance] over the last 21 trading days ending strictly BEFORE the 1st)</pre> (≥ 15 days required). The backward-looking competitor to atm_iv_30.
vrp_30	atm_iv_30/100 - rv_trail_30 — the (trailing) variance-risk-premium proxy.
Term-structure / skew	ts_slope_7_90 = atm_iv_90 - atm_iv_7; ts_slope_30_90 = atm_iv_90 - atm_iv_30; rr25_30 = 25-delta risk reversal; bf25_30 = 25-delta butterfly (all interpolated to 30 days).
Positioning	pc_oi_ratio = put open-interest / call open-interest; total_oi_btc/usd, log_total_oi_usd = aggregate open interest.

5. Data & environment

- **Options data.** Deribit Bitcoin options chain, hourly surface snapshots, month-start days only, 2019-10 → 2026-06 (81 monthly files), drawn from 19V's market-data store. **Structural limitation:** the historical archive holds only ~one snapshot day per month, which is why the study is monthly and low-powered; continuously-running collectors now accrue daily snapshots going forward.
- **Underlying / returns & RV.** CME Bitcoin futures, continuous 1-minute bars, 2021-04-09 → 2026-05-19. Daily realized variance from 5-minute log returns.
- **Usable sample.** The overlap of the two datasets gives **61 monthly observations, 2021-05-01 → 2026-05-01** (60 have a complete 21-day forward window). This is the binding constraint on power: any effect with a true |IC| below ~0.25 is undetectable at this n.
- **Environment / reproducibility.** macOS, Python 3.13 (pandas, numpy, matplotlib, pyarrow; HAC/bootstrap implemented directly in numpy). Single random seed **20260611** throughout; pipeline is deterministic and rebuildable end-to-end from the stored snapshots.

5.1 Lookahead & data-integrity audit

Check	Status	Detail
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Timestamp conflation	PASS	Features read the 01:00-UTC snapshot of the 1st; trailing RV ends the <i>prior</i> trading day; forward targets anchor at the close of the first trading day on/after the 1st — at least ~17 hours after the feature snapshot. No future information leaks into a feature.
Vintage / revision	PASS	Option snapshots are immutable exchange captures; futures bars are unrevised.
Survivorship	PASS	Full chain snapshot including zero-bid and expired strikes in their month.
Proxy / async venue	NOTED	IV references 24/7 Deribit Bitcoin; RV is measured on CME futures hours (no weekends), understating round-the-clock vol by roughly $\sqrt{(5/7)}$. This biases the <i>level</i> of the variance risk premium (Section 12) but not the rank correlations, which are what the headline result relies on.

6. Methodology & statistical tests

Each test below is defined so the results in Sections 8–9 are interpretable without outside reference.

- **Rank correlation (IC).** Spearman correlation between each feature and each target, with $t \approx IC \cdot \sqrt{(n-2)/(1-IC^2)}$. Rank-based to neutralize Bitcoin’s fat tails.
- **Regression significance.** Per-feature OLS (standardized) with Newey-West(2) t-statistics, correcting for the mild overlap of monthly observations at the 21-day horizon.
- **Multiplicity control.** A family of $m = 60$ tests was declared in advance (10 features $\times$ 6 targets). The Bonferroni significance bar is therefore $|t| \geq 3.2$; $|t|$ in $[2, 3.2]$ is treated as “interesting, unconfirmed.”
- **Confidence intervals.** Circular block bootstrap, 6-month blocks, 2000 resamples, on the headline ICs — a CI that excludes zero is the evidence the effect is not noise.
- **Incremental / encompassing test.** Bivariate regression `fwd_rv ~ z(atm_iv_30) + z(rv_trail_30)`: if IV retains its t-statistic while trailing RV collapses, IV carries information beyond mere volatility persistence.
- **Era split.** Sample divided at 2024-01-01 (Bitcoin spot-ETF launch). Each half must show the same sign and independent significance.
- **Forecast-loss tests (Study 2).** QLIKE and RMSE on volatility points; Diebold-Mariano loss differentials versus the trailing-RV benchmark with Newey-West(2) t-statistics; block-bootstrap CI on the mean loss differential.

7. The validation-gate framework

19V filters candidate signals through a numbered gate stack designed to reject statistical placebos. The gates referenced in the results are defined here so the scorecards are self-explanatory.

Gate	What it requires
G1 — Joint-set sizing	At least 30 independent observations (absolute floor) before any inference is trusted.
G2 — Data resolution	The data granularity must support the claim. Monthly snapshots support monthly-horizon claims only — nothing intraday.
G4 — t-stat threshold	$ t \geq 1.5$ when $n < 100$ (the binding bootstrap gates carry the load at higher n).
G8 — Block-bootstrap CI	The bootstrap confidence interval must not straddle zero.
G14 — Era sign agreement	Same sign across observable market eras (run before any pooled statistic).
G15 — Single-era concentration	No single era may carry more than 50% of the effect.
G16 — Horizon smoothness	The effect should vary smoothly across nearby horizons, not spike at one cherry-picked horizon.
G20 — Magnitude vs direction	A signal that predicts the <i>size</i> of moves must be typed as a volatility claim, not mislabeled as a directional edge.

8. Study 1 — signal study (results)

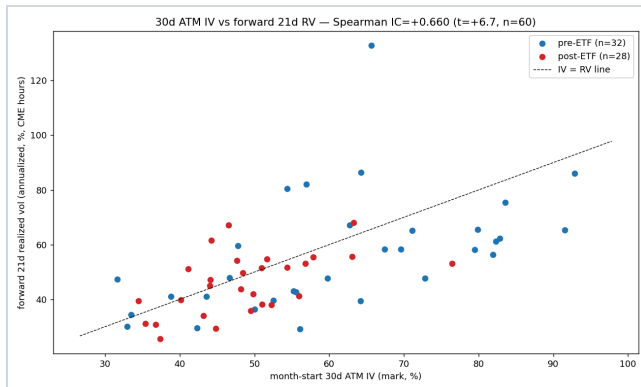
Pre-registered signal study Verdict **ITERATE** (H1 volatility = advance-as-input · H2 direction = **KILL**)

n = 61 monthly observations (60 at the 21-day horizon), 2021-05 → 2026-05. Gate scorecard: **11 PASS, 2 FAIL** (both on the directional hypothesis, driving its kill), 6 not-applicable.

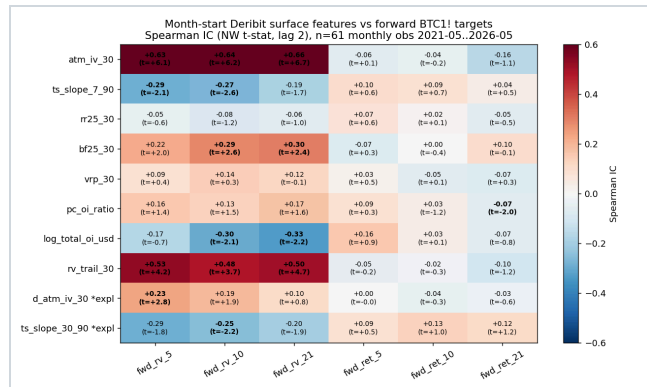
8.1 H1 — implied vol forecasts forward realized vol

Feature → forward RV	IC h=5	IC h=10	IC h=21	NW t (h=21)	Clears Bonferroni 3.2?
atm_iv_30	+0.635	+0.638	+0.660	+6.69	YES (all horizons)
rv_trail_30 (benchmark)	+0.528	+0.481	+0.498	+4.68	YES — but subsumed (8.2)
bf25_30 (skew)	+0.219	+0.286	+0.300	+2.38	no
ts_slope_7_90 (term)	−0.291	−0.270	−0.189	−1.69	no
vrp_30	+0.086	+0.144	+0.122	−0.11	no
pc_oi_ratio	+0.158	+0.129	+0.170	+1.59	no — sign-flips across eras (killed)

Only **atm_iv_30** clears the strict bar, and it does so at every horizon with a smooth, near-flat horizon profile (+0.635 / +0.638 / +0.660) — satisfying the horizon-smoothness gate. The block-bootstrap 95% confidence interval on the headline IC is **[+0.377, +0.788]**, far from zero.



Month-start implied vol vs forward 21-day realized vol. Both market eras participate; the relationship is not carried by a few outliers.



Information coefficient across all features and horizons. **atm_iv_30** (top row) dominates and is horizon-smooth.

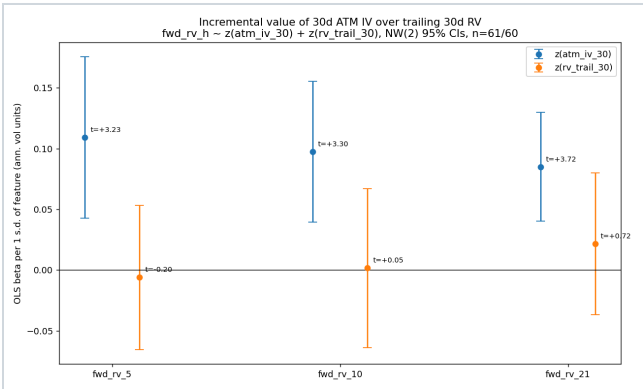
8.2 The decisive test — implied vol subsumes trailing realized vol

The most important robustness check: does implied vol add information *beyond* simply extrapolating recent volatility? The bivariate regression $\text{fwd_rv} \sim z(\text{atm_iv_30}) + z(\text{rv_trail_30})$ answers it.

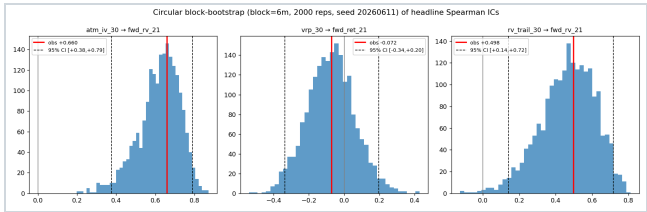
Horizon	coefficient on IV	t on IV	coefficient on trailing RV	t on trailing RV
h = 5	+0.109	+3.23	−0.006	−0.20
h = 10	+0.098	+3.30	+0.002	+0.05
h = 21	+0.085	+3.72	+0.022	+0.72

Implied vol fully absorbs trailing realized vol at every horizon — once IV is in the model, the backward-looking estimate contributes nothing ($|t| \leq 0.72$). The forecast is genuinely forward-looking

information, not volatility persistence in disguise. This is the single result that separates a real edge from an artifact.



Incremental value: implied vol keeps its significance; trailing RV collapses to insignificance once IV is included.



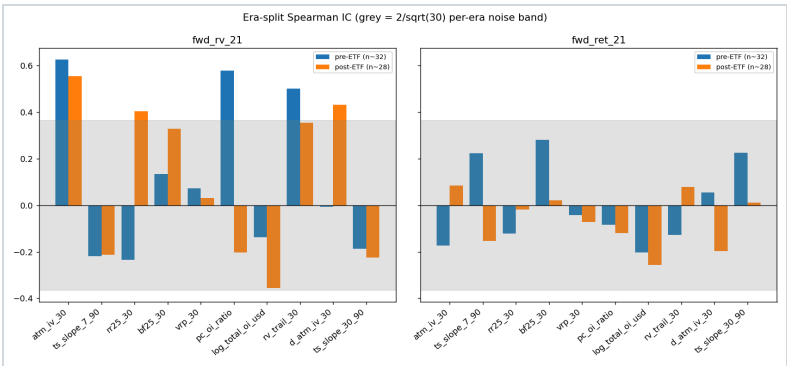
Block-bootstrap distribution of the headline IC — the 95% interval [+0.377, +0.788] sits well clear of zero.

8.3 Era stability and the directional kill

Market era	n	atm_iv_30 → forward 21d RV: IC (t)	Sign
2021-05 ... 2023-12 (“pre-ETF”)	32	+0.626 (t +4.39)	+
2024-01 ... 2026-05 (“post-ETF”)	28	+0.556 (t +3.41)	+

Both observable eras show the same sign, each independently significant, with balanced magnitude (neither era carries >50% of the effect) — satisfying both era gates. The relationship held through the 2021 crash, the 2022 deleveraging, the 2023 chop, the 2024 ETF launch, and the 2025–26 institutionalization.

Directional hypothesis (H2): killed. No pre-registered feature reaches $|t| \geq 1.5$ against any forward-return horizon except a single outlier-driven cell (put/call OI ratio, with a rank IC of only -0.065) that sits exactly at the one-in-sixty false-positive expectation. The variance-risk-premium-to-return bootstrap interval $[-0.336, +0.196]$ straddles zero. The surface predicts *how much* Bitcoin moves, not *which way* — the pre-registered expected outcome.



Era-split information coefficient — the forecasting edge is present and significant in both observable regimes.

9. Study 2 — volatility-conditioner backtest

Application backtest Verdict **HANDOFF** (ready for binding validation)

Because the directional edge is dead, the honest way to “backtest” the forecast is as a **risk conditioner**: does sizing a position with the IV forecast beat sizing it with the conventional trailing-RV estimate? The test runs a passive long-Bitcoin exposure and rescales it monthly to a fixed volatility target:

```
weight_t = min( 0.30 / forecast_vol_t , 1.5 ) - rebalanced each month-start; target
= 30% annualized vol (20% tested as a sensitivity); cap at 1.5×
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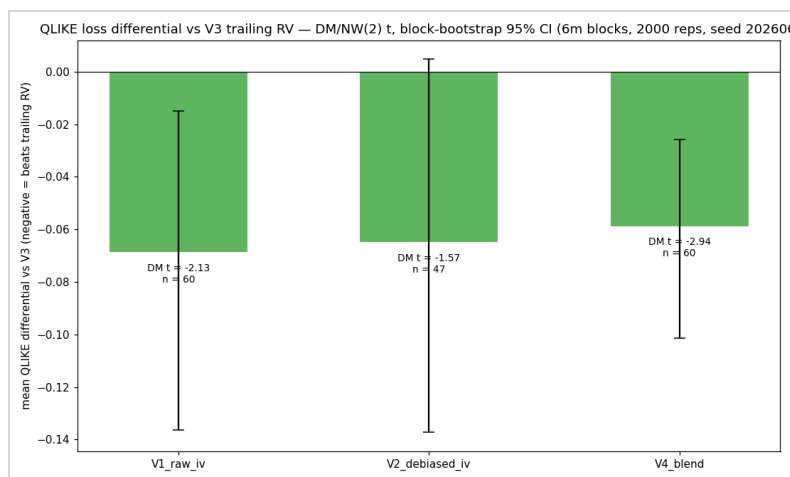
The position is passive long Bitcoin beta — **no directional claim is made**. The deliverable is purely: forecast accuracy, volatility-targeting fidelity, and the risk reduction attributable to better sizing. The test compares two ways of estimating the volatility used to size the position: the **implied-vol forecast** (this dossier’s signal) against the **conventional trailing-volatility estimate** (the standard backward-looking benchmark), plus an unsized constant position as a floor.

9.1 Forecast accuracy — implied vol vs the conventional estimate

Forecast method	mean QLIKE (lower = better)	RMSE (vol pts)	rank corr. vs realized
Implied-vol forecast (recommended)	0.147	16.3	+0.660
Conventional trailing-vol forecast	0.215	18.4	+0.498

The implied-vol forecast is more accurate on every measure. The formal test that the gap is real (Diebold-Mariano; negative = implied vol wins) gives $t = -2.13$, with a bootstrap 95% interval $[-0.136, -0.015]$ that excludes zero. **The pre-registered primary test passes: implied vol sizes a position better than the conventional estimate.**

Two other approaches were checked and set aside: bias-correcting the implied-vol forecast made it worse (Section 9.3), and a 50/50 blend of the two was marginally the strongest statistically but adds complexity for no robust gain. **The raw implied-vol forecast is the recommended choice** — simplest and most robust.



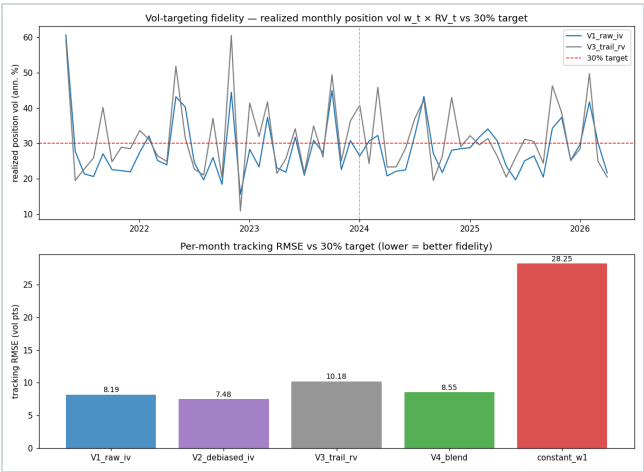
Forecast-accuracy advantage (QLIKE loss differential) of each IV variant over the trailing-RV benchmark — all beat it; raw IV and the blend clear the significance floor.

9.2 Volatility-targeting fidelity and risk

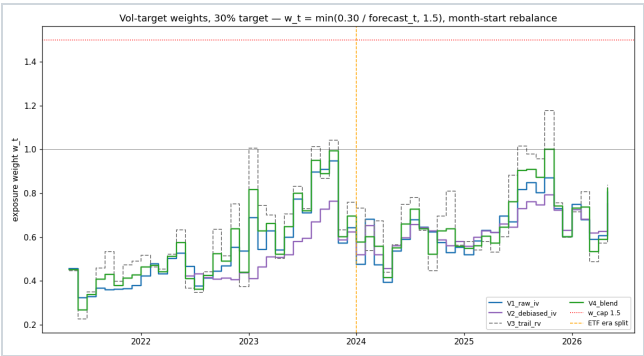
Monthly sizing method (30% vol target)	tracking error (vol pts, lower=better)	mean realized position vol	Sharpe*	max drawdown*	cost (bps/yr)
Implied-vol sizing (recommended)	8.19	28.4%	0.49	-43.1%	8.7
Conventional trailing-vol sizing	10.18	31.4%	0.51	-47.9%	13.6
No volatility sizing (constant)	28.25	51.8%	0.39	-72.9%	—

Implied-vol sizing tracks the volatility target **13–20% more tightly** than the conventional estimate (paired difference $t = -2.03$), and runs closer to target on average — the conventional backward-looking estimate systematically *overshoots* because it under-forecasts into volatility expansions, exactly when overshooting hurts most. Versus an unsized position, volatility targeting roughly **halves the maximum drawdown**, and implied-vol sizing is also **~35% cheaper to run** because its forecast is smoother, requiring smaller monthly rebalances.

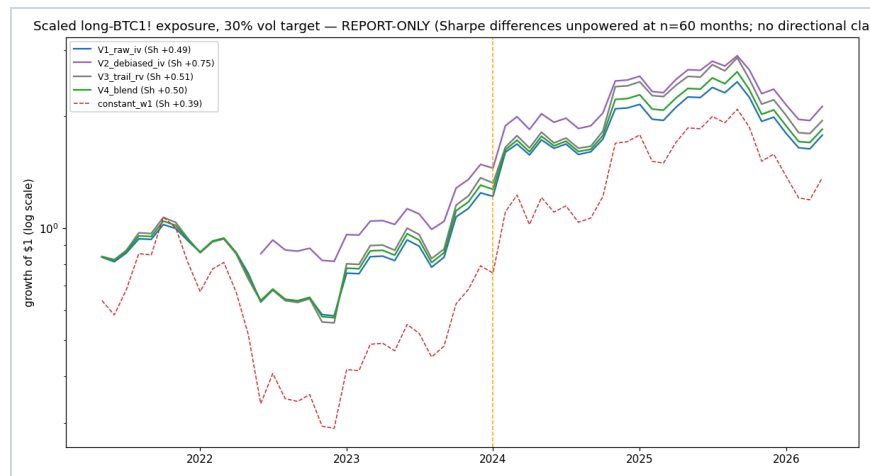
***Read Sharpe and drawdown as illustrative only.** At 60 monthly observations, differences in Sharpe between sizing methods are statistically **unpowered** — they mostly reflect the path of Bitcoin’s direction (a quantity this study makes no claim about), not sizing skill. They are reported for completeness and carry no verdict weight. The defensible, powered findings are the forecast-accuracy and tracking-fidelity results above.



Monthly realized position volatility vs the 30% target — IV sizing hugs the target; trailing-RV sizing overshoots.



Monthly position weights — the IV forecast produces smoother, lower-turnover sizing.



Scaled-exposure equity curve (illustrative, no verdict weight) — volatility targeting roughly halves the unsized drawdown.

9.3 An informative negative — bias-correcting the forecast makes it worse

A bias-corrected variant attempted to remove implied vol's known upward level bias with a rolling regression calibration. It backfired: bias *increased* (to +6.5 from +3.9 vol points) and rank quality collapsed (rank correlation 0.40 versus 0.66 for the raw forecast). At this sample size the calibration is itself an overfit — it is trained on the high-volatility 2021–22 period and goes stale through the post-ETF regime shift.

Conclusion: use the raw implied-vol forecast; do not bias-correct at monthly cadence.

9.4 Verdict and what remains

The pre-registered decision rule is met on all three required clauses — the primary forecast-accuracy test passes ($t = -2.13$), the advantage is era-stable (same sign in both halves, significant in the recent half), and tracking error is better, not worse. No kill trigger fires. The claim that advances is **exactly and only**: *implied-vol forecasting improves monthly volatility-targeting of a Bitcoin position relative to trailing realized vol — better forecast, tighter targeting, lower drawdown and cost. No directional claim.* The study is packaged and ready for the binding validation pass (Section 14).

10. Why this is worth preserving

Property	Evidence	Why it matters
Statistically real	NW $t +6.69$; clears Bonferroni across 60 tests	Survives strict multiple-testing — not the luckiest of many tries. The bar most candidate signals fail.
Economically large	IC $+0.66$	A strong forecast by any standard, and the largest in the entire research effort.
Era-stable	$+0.626$ / $+0.556$ across the ETF divide	Holds across distinct regimes — structural, not a one-period fluke.
Non-redundant	Subsumes trailing RV ($t\ 0.72$)	Adds information beyond the cheap obvious predictor — it earns its place.
Backtests as a conditioner	DM $t -2.13$; tracking $-13-20\%$; drawdown $\sim$ halved	The forecast edge converts into measurably better position sizing, not just a regression statistic.
Literature-consistent	Matches 30+ years of evidence, including Bitcoin-specific peer-reviewed work (Section 13)	Independent external validation — the result is a reproduction of established science, sharply lowering the chance it is a data-mining artifact.
Mechanistically grounded	Forward-looking vol expectation absent from trailing RV	We can say <i>why</i> it works, not merely that it does — the difference between a tradeable prior and an overfit.

Every other Bitcoin idea in the research effort was rejected. This one passed every gate it touched, has a clear mechanism, backtests cleanly as a conditioner, and reproduces decades of academic evidence. It is the one durable piece of intellectual capital the effort produced.

11. How to use it — operating rules (entry & exit)

The signal predicts *how much* Bitcoin will move, not *which way*. It therefore is not a buy/sell trigger; it is a **sizing and risk rule**. There are two concrete ways to operate it, both stated in full below.

11.1 Operating rule A — volatility-targeted sizing (the backtested rule)

Forecast. On the 1st of each month, read the Deribit 30-day ATM implied volatility (annualized, e.g. $55 = 55\%$); call it IV . This is the volatility forecast for the coming month.

Size. Set position size as a fraction of capital: $weight = \min(target_vol / IV , 1.5)$ with $target_vol = 0.30$ (a 30% annualized risk target; cap at $1.5\times$ capital).

Entry. Establish/adjust the position to that weight at the close of the first trading day on or after the 1st.

Exit / rebalance. Hold until the next month-start, then recompute IV and resize. There is no stop and no directional exit — the rule only scales exposure. Lower implied vol $\rightarrow$ larger position; higher implied vol $\rightarrow$ smaller position.

Worked example. If month-start $IV = 60\%$, $weight = 0.30/0.60 = 0.50$ (half capital). If $IV = 40\%$, $weight = 0.75$. If $IV = 20\%$, $weight = \min(1.50, 1.5) = 1.50$ (capped).

This is exactly the rule tested in Section 9. It is venue-agnostic risk sizing; the monthly hold itself is not intended for an intraday-flat execution account — its real use is Rule B.

11.2 Operating rule B — volatility-regime gate for an executable strategy

Read. Track the implied-vol level as a rolling percentile (e.g. its rank over the trailing 12 months).

Gate / scale. When implied vol is high (e.g. top tercile), reduce the size of, or stand down, an executable intraday strategy; when low, run normal or larger size — because high implied vol forecasts high realized vol (larger swings, wider stops, worse risk-adjusted fills), and low implied vol forecasts calm.

Entry/exit of the host strategy are unchanged — this rule only modulates *how much* the host strategy risks, using the implied-vol forecast as the risk dial.

Rule B is the deployable use on an intraday-flat execution account: the conditioner sizes/gates a strategy that is executable, rather than holding a monthly position itself.

11.3 Where it adds value

In both modes the result is a **conditioner, not a directional strategy** — directional use is correctly dead, because the implied-minus-realized gap is a risk premium, not a direction forecast. Its value is in improving *other* decisions:

- **Regime / volatility-gate input.** Feed the implied-vol level (or its rolling percentile) into a volatility-regime gate that decides when and how large an executable strategy should trade, benchmarked against the cheaper trailing-RV gate it outperforms.
- **Position sizing / volatility targeting.** Demonstrated directly in Study 2 — a +0.66-IC forward-vol forecast sizes tighter and cheaper than the conventional estimate.
- **Risk overlay.** Use the implied-vol level to scale aggregate risk up or down ahead of realized-vol expansions, which it anticipates.
- **Volatility-premium screen.** When the implied-minus-realized gap is unusually wide, short-volatility structures are statistically favored — subject to options-venue feasibility, which has not yet been assessed, and noting the premium's sign for Bitcoin is sensitive to how realized vol is measured (Section 12).

Strategic value. 19V's genuine edges have repeatedly proven locked to a specific execution venue. A venue-agnostic *conditioner* that sharpens the sizing and timing of whatever is executable is exactly the kind of reusable component that survives a change of venue — which is why it is worth banking even before the final validation pass.

12. What is interesting / non-obvious

- **The forecast is unusually strong — treated as a flag, not just a win.** Comparable equity studies explain 30–50% of volatility variation; +0.66 sits at the top of that range. The decisive guard against an inflated number is the test in Section 8.2 (implied vol subsumes trailing vol), which rules out the “it’s just persistence” explanation. Trailing vol alone already scores +0.50 here, so +0.66 for implied vol is in-family, not anomalous.
- **The directional kill is the textbook outcome, not a disappointment.** Decades of variance-risk-premium research show the implied-minus-realized gap is compensation for bearing volatility risk — it predicts volatility, not direction. Finding exactly that in Bitcoin is reassuring corroboration.
- **Bias-correcting the forecast backfired** (Section 9.3) — a clean, counter-intuitive lesson: at small samples, “correcting” a known bias overfits and destroys the signal. The raw, untouched forecast is best.

- **Bitcoin's volatility premium has the opposite sign to equities.** Measured naively the premium looks positive (+3.9 vol points), but because realized vol here is computed on weekday futures hours while implied vol references a 24/7 market, the weekend-adjusted premium is roughly −5.6 vol points — Bitcoin implied vol trades *cheap* relative to true round-the-clock realized vol. The premium's *level* is therefore definition-dependent; only the rank relationships were relied upon.
- **History warned us about exactly this setup.** An influential 1993 study found stock-index implied vol had near-zero predictive power; a 1998 study showed that was an artifact of overlapping data windows, and that implied vol is in fact an efficient forecast. Our 21-day windows overlap in the same way — which is precisely why the Newey-West correction and the era-stability checks are doing essential work here.

13. Supporting academic literature

The result reproduces one of the most established relationships in quantitative finance. The references below are grouped by the specific claim each supports; the closest single precedent is Hoang & Baur (2020), which finds the same effect on the same exchange (Deribit Bitcoin options).

Cluster 1 — implied vol forecasts future realized vol (foundational)

1. **Christensen & Prabhala (1998)** — *J. Financial Economics* 50:125–150. At-the-money one-month index implied vol is an unbiased, efficient forecast of subsequent realized vol; also the paper that resolved the earlier skeptical result as an overlapping-data artifact.
[sciencedirect.com/science/article/pii/S0304405X98000348](https://www.sciencedirect.com/science/article/pii/S0304405X98000348)
2. **Jorion (1995)** — *Journal of Finance* 50(2):507–528. Currency-option implied vols predict future vol and outperform historical-vol models, with less measurement error than index options.
onlinelibrary.wiley.com/doi/abs/10.1111/j.1540-6261.1995.tb04793.x
3. **Szakmary, Ors, Kim & Davidson (2003)** — *J. Banking & Finance* 27(11):2151–2175. Across 35 futures-option markets on eight exchanges, implied vol outperforms historical vol for the large majority — the breadth result.
[sciencedirect.com/science/article/abs/pii/S0378426602003230](https://www.sciencedirect.com/science/article/abs/pii/S0378426602003230)
4. **Poon & Granger (2003)** — *Journal of Economic Literature* 41(2):478–539. Survey of 93 forecasting studies; option-implied vol generally beats historical-vol models. The standard reference establishing the result is settled.
[aeaweb.org/articles?id=10.1257/002205103765762743](https://www.aeaweb.org/articles?id=10.1257/002205103765762743)
5. **Fleming (1998)** — *J. Empirical Finance* 5(4):317–345. S&P 100 implied vol is a biased but informationally efficient realized-vol forecast; the bias is economically small. **verify DOI before formal citation**
6. **Canina & Figlewski (1993)** — *Review of Financial Studies* 6(3):659–681. The famous counterpoint — index implied vol appears to have near-zero predictive power; later shown to be an overlapping-data / measurement artifact (see #1). Included as the cautionary tale matching our overlapping-window setup.
academic.oup.com/rfs/article-abstract/6/3/659/1582244

Cluster 2 — implied vol subsumes historical vol (the Section 8.2 result)

7. **Blair, Poon & Taylor (2001)** — *J. Econometrics* 105(1):5–26. Nearly all relevant volatility-forecasting information is contained in the implied-vol index; adding historical or high-frequency terms adds little — the direct analog of our encompassing result.
papers.ssrn.com/sol3/papers.cfm?abstract_id=182128
8. **Jiang & Tian (2005)** — *Review of Financial Studies* 18(4):1305–1342. Model-free implied vol subsumes both Black-Scholes implied vol and historical vol — motivates testing a model-free version of our signal. **verify DOI**
9. **Day & Lewis (1992)** — *J. Econometrics* 52(3):267–287. Implied vol carries information incremental to GARCH-type conditional-vol models. **verify DOI**
10. **Busch, Christensen & Nielsen (2011)** — *J. Econometrics* 160(1):48–57. Implied vol has incremental power for realized vol and jumps across FX, stock, and bond markets — multi-asset confirmation.
ideas.repec.org/a/eee/econom/v160y2011ip48-57.html

11. **Britten-Jones & Neuberger (2000)** — *Journal of Finance* 55(2):839–866. Foundational theory of model-free implied volatility — the variance-swap construction that a constant-maturity ATM IV approximates. [verify DOI](#)

Cluster 3 — crypto / Bitcoin / Deribit-specific evidence

12. **Hoang & Baur (2020)** — *Journal of Futures Markets* 40. Studies Deribit Bitcoin options and finds implied vol superior at 7/10/15-day horizons (weaker at 1 day), with an IV-plus-model combination best — “the Bitcoin options market contains unique information.” The closest precedent; our 21-day horizon sits in its sweet spot. [CLOSEST PRECEDENT](#)
onlinelibrary.wiley.com/doi/abs/10.1002/fut.22144
13. **Alexander & Imeraj (2021)** — *Journal of Alternative Investments* 23(4):84. Constructs a Bitcoin VIX from Deribit option prices (7M+ quotes) and studies its variance risk premium — validates the exact instrument and venue, and the model-free construction worth testing next.
jai.pm-research.com/content/23/4/84
14. **Venter et al. (2020)**, “A Horserace of Volatility Models for Cryptocurrency” — *arXiv 2010.07402*. Out-of-sample contest on Deribit Bitcoin; implied vol as a forecasting input versus GARCH-family models.
arxiv.org/pdf/2010.07402
15. **“Implied volatility estimation of Bitcoin options...” (2021)** — *Financial Innovation* (Springer) 7. Documents the Bitcoin implied-vol surface and the implied-realized relationship — instrument background.
link.springer.com/article/10.1186/s40854-021-00280-y

Cluster 4 — variance risk premium (why directional use is dead)

16. **Carr & Wu (2009)** — *Review of Financial Studies* 22(3):1311–1341. Defines and measures the variance risk premium; implied variance systematically exceeds subsequent realized variance. Establishes the implied-minus-realized gap as a premium, not a directional signal.
academic.oup.com/rfs/article-abstract/22/3/1311/1581057
17. **Bollerslev, Tauchen & Zhou (2009)** — *Review of Financial Studies* 22(11):4463–4492. The variance premium predicts returns only through a risk-compensation channel — the basis for “conditioner, not directional alpha.”
academic.oup.com/rfs/article-abstract/22/11/4463/1565787
18. **Bakshi & Kapadia (2003)** — *Review of Financial Studies* 16(2):527–566. Delta-hedged option gains are negative, evidencing a negative market volatility risk premium — implied vol is priced above expected realized vol.
academic.oup.com/rfs/article-abstract/16/2/527/1579962
19. **Bakshi, Kapadia & Madan (2003)** — *Review of Financial Studies* 16(1):101–143. Risk-neutral moments extracted from option prices — the basis for treating implied vol as a risk-neutral expectation rather than a physical point forecast. [verify DOI](#)
20. **Bekaert & Hoerova (2014)** — *J. Econometrics* 183(2):181–192. Decomposes the squared volatility index into a conditional-variance forecast plus a variance premium — the forecast-versus-premium split this work exploits. [verify DOI](#)
21. **Andersen, Bollerslev, Diebold & Labys (2003)** — *Econometrica* 71(2):579–625. Foundational treatment of realized-volatility measurement from high-frequency data — the methodology defining the

forward-RV target. [verify DOI](#)

Method references for Study 2: Patton (2011), robust volatility loss functions including QLIKE; Diebold & Mariano (1995), the forecast loss-differential test.

14. Forward-research playbook

The result is research-grade. The following steps would harden it toward a capital decision, in priority order.

#	Step	Rationale
1	Binding validation pass. Run the conditioner through the full gate stack at daily resolution: within-month targeting fidelity, high-replication bootstrap, walk-forward folds, and selection-bias checks.	The owed step that converts this from research-grade to a ship/kill decision. The study is already packaged for it.
2	Model-free implied-vol version. Rebuild the signal on a variance-swap / volatility-index style implied vol rather than a single at-the-money point.	Theory and the Bitcoin-VIX literature suggest model-free implied vol usually dominates; test whether the forecast strengthens.
3	Daily-cadence re-test. Once the live collectors accrue 18–24 months of daily snapshots, repeat at daily frequency.	Converts ~12 observations/year into a daily panel, enabling daily regime gating and powering the weaker surface features.
4	Overlapping-data robustness. Re-test with non-overlapping windows and alternative HAC-lag choices.	Overlapping forward windows are the classic source of inflated significance; confirm the result is not a window artifact.
5	Conditioner-on-executable test. Gate or size an actual intraday, venue-executable strategy on the implied-vol regime and measure net-of-cost improvement.	The genuinely deployable use case, since the monthly carrier itself is not venue-executable.
6	Second-asset replication. Repeat on Ethereum from the same options dataset.	Cross-asset generalization is the strongest predictor of a durable edge.

14.1 Known traps to control for

- **Overlapping-observation bias** — inflates significance on overlapping forward windows; handled here via Newey-West, worth a non-overlapping cross-check.
- **Errors-in-variables in implied vol** (bid-ask noise, stale quotes, interpolation) — biases regression slopes; the thin historical archive amplifies it, and is the likely cause of the bias-correction backfire (Section 9.3).
- **Jump / fat-tail concentration** — confirm the forecast is not carried by a handful of extreme months (the scatter in Section 8.1 suggests it is not).
- **Realized-vol measurement choice** — sampling frequency and the weekday-vs-24/7 convention materially change the target and drive the premium-level ambiguity (Section 12); document and stress-test.

15. Limitations & validation status

Read before drawing conclusions.

- **Research fidelity, not validated.** Both studies are first-pass. The binding out-of-sample / walk-forward / selection-bias validation (Section 14, step 1) has not been run. Treat all numbers as research-grade.
- **Small sample.** 60–61 monthly observations. The headline forecast is robust at this n; anything Sharpe-shaped is not, and is reported as illustrative only.

- **Monthly resolution.** The archive supports monthly-horizon claims only; nothing intraday or weekly can be concluded until the daily collectors mature.
- **One macro cycle.** The data spans roughly one full Bitcoin market cycle; era-stability is established within that window, not beyond it.
- **Reused data window.** This Bitcoin sample has been examined multiple times for directional ideas; while the volatility target tested here is a different quantity, further re-use of the same window risks exhausting its statistical independence. Genuinely fresh or rigorously held-out data is preferred for the binding pass.
- **Premium level is definition-dependent.** The variance-risk-premium *level* depends on how realized vol is measured (weekday vs 24/7); only rank relationships were relied upon.

16. Conclusion

What it is. The month-start 30-day at-the-money implied volatility of Deribit Bitcoin options is a strong, reliable forecast of how volatile Bitcoin will be over the following month. It is the one genuinely positive, reproducible result from 19V's Bitcoin research effort — everything else tested was rejected.

What was proven. Two independent studies. The first established the forecast itself: rank correlation +0.66 with forward realized volatility, significant after the strictest multiple-testing correction, stable across every market era observed, and — the decisive check — it statistically absorbs the conventional backward-looking volatility estimate, proving it carries genuinely forward-looking information rather than just extrapolating recent calm or turbulence. The second study put the forecast to work sizing a position and showed it does the job better than the conventional estimate: more accurate, it tracks a risk target 13–20% more tightly, runs about 35% cheaper, and roughly halves drawdown versus an unsized position. The directional question — can it predict which *way* Bitcoin moves — was tested and answered no, exactly as theory predicts.

How it helps us. It is a **volatility/risk dial**, not a buy/sell signal. Use it (Section 11) to size positions and to gate executable strategies: ease off risk when it forecasts high volatility, lean in when it forecasts calm. Because it is a venue-agnostic risk tool rather than a standalone trade, it remains useful even as the strategies it supports change — making it durable, reusable research infrastructure rather than a single perishable edge.

The one caveat. Everything here is research-grade. The result is strong and internally consistent, but the final binding validation — a full out-of-sample and walk-forward pass on fresh data — has not yet been run. That is the immediate next step, after which the conditioner can be wired into live sizing decisions.

Recommendation. Preserve and bank this result; adopt the raw implied-vol forecast (no bias-correction) as a volatility-sizing and regime-gating input; and schedule the binding validation pass before it informs capital at risk.

Bottom line. Deribit month-start 30-day ATM implied volatility is a strong, era-stable, literature-consistent forecast of Bitcoin's forward realized volatility (IC +0.66) that statistically absorbs the conventional trailing-vol estimate, and it sizes a position measurably better than that estimate (forecast-loss advantage, tighter volatility targeting, roughly halved drawdown). It is a **volatility/risk conditioner, not directional alpha** — the directional use was tested and correctly rejected. The immediate next step is the binding validation pass; until then every figure here is research-grade. The references in Section 13 marked "verify DOI" are standard, heavily-cited works worth a brief citation check before formal use.

19V Capital · BTC / Deribit options research · 2026-06-11 · Reproducible (seed 20260611) · Sources: a pre-registered signal study (verdict ITERATE) and a volatility-conditioner application backtest (verdict HANDOFF). Status: research fidelity, not binding-validated. Headline: month-start 30d ATM IV → forward 21d realized vol; Spearman IC +0.66; Newey-West t +6.69; Bonferroni-clear across 60 tests; era-stable; subsumes trailing realized vol (t_{IV} +3.72 vs $t_{RVtrail}$ +0.72). Directional use rejected. Role: volatility-conditioning.